

Some Classical Attacks

Shibam Ghosh¹

June 7, 2026

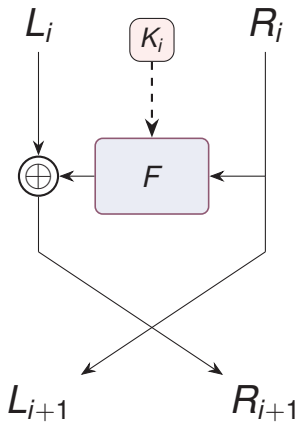
¹Department of CSE, IIT Hyderabad

Table of contents

1. The Round Function of a Feistel Structure
2. Meet-in-the-middle

The Round Function of a Feistel Structure

The Feistel Structure

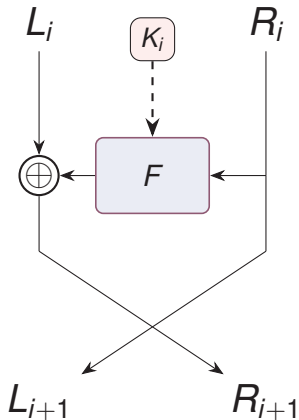


Encryption Round

$$L_{i+1} = R_i$$

$$R_{i+1} = L_i \oplus F(K_i, R_i)$$

Exploring Weak Functions



Case 1: The Zero Function

What if $F(K, x) = 0 \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus 0 = L_0$$

$$L_2 = L_0$$

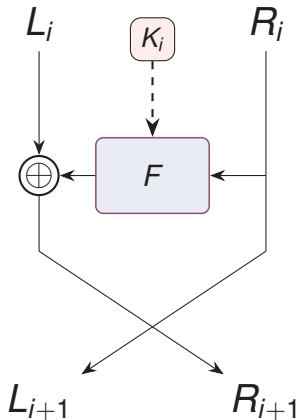
$$R_2 = R_0$$

Repeat

After n rounds, $L_n = L_0$ and $R_n = R_0$

Weak function

Exploring Weak Functions



Case 1: The Zero Function

What if $F(K, x) = 0 \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus 0 = L_0$$

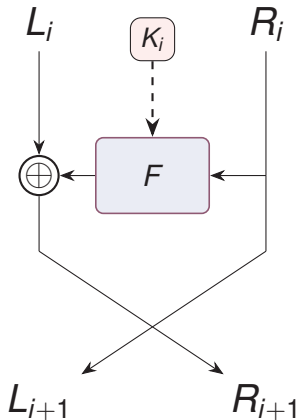
$$L_2 = L_0$$

$$R_2 = R_0$$

Result

After an even number of rounds, you get the input back. **No security!**

Exploring Weak Functions



Case 1: The Zero Function

What if $F(K, x) = 0 \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus 0 = L_0$$

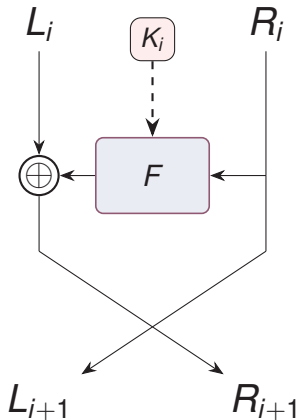
$$L_2 = L_0$$

$$R_2 = R_0$$

Result

After an even number of rounds, you get the input back. **No security!**

Exploring Weak Functions



Case 1: The Zero Function

What if $F(K, x) = 0 \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus 0 = L_0$$

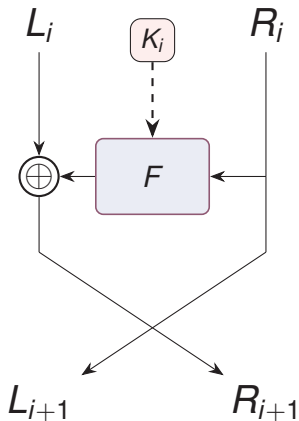
$$L_2 = L_0$$

$$R_2 = R_0$$

Result

After an even number of rounds, you get the input back. **No security!**

Exploring Weak Functions



Case 2: The Identity Function

What if $F(K, x) = x, \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0$$

$$L_2 = L_0 \oplus R_0$$

$$R_2 = R_0 \oplus (L_0 \oplus R_0) = L_0$$

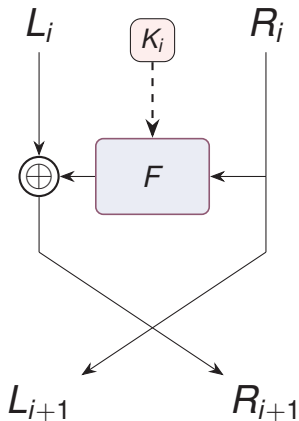
$$L_3 = L_0$$

$$R_3 = (L_0 \oplus R_0) \oplus L_0 = R_0$$

Result

The input (L_0, R_0) is restored after just 3 rounds. **No security!**

Exploring Weak Functions



Case 2: The Identity Function

What if $F(K, x) = x, \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0$$

$$L_2 = L_0 \oplus R_0$$

$$R_2 = R_0 \oplus (L_0 \oplus R_0) = L_0$$

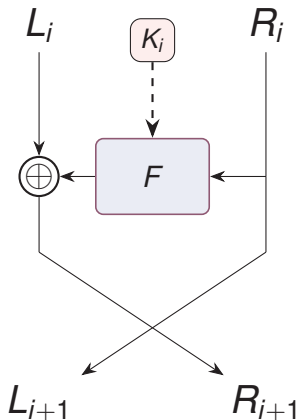
$$L_3 = L_0$$

$$R_3 = (L_0 \oplus R_0) \oplus L_0 = R_0$$

Result

The input (L_0, R_0) is restored after just 3 rounds. **No security!**

Exploring Weak Functions



Case 2: The Identity Function

What if $F(K, x) = x, \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0$$

$$L_2 = L_0 \oplus R_0$$

$$R_2 = R_0 \oplus (L_0 \oplus R_0) = L_0$$

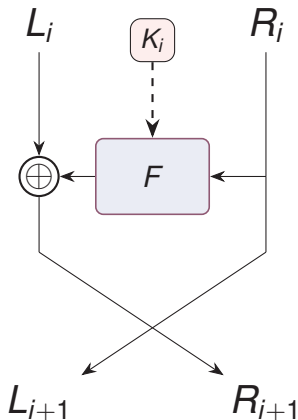
$$L_3 = L_0$$

$$R_3 = (L_0 \oplus R_0) \oplus L_0 = R_0$$

Result

The input (L_0, R_0) is restored after just 3 rounds. **No security!**

Exploring Weak Functions



Case 2: The Identity Function

What if $F(K, x) = x, \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0$$

$$L_2 = L_0 \oplus R_0$$

$$R_2 = R_0 \oplus (L_0 \oplus R_0) = L_0$$

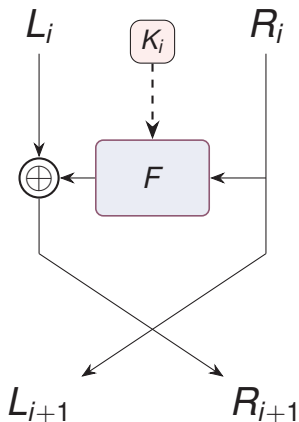
$$L_3 = L_0$$

$$R_3 = (L_0 \oplus R_0) \oplus L_0 = R_0$$

Result

The input (L_0, R_0) is restored after just 3 rounds. **No security!**

Exploring Weak Functions



Case 3: Simple XOR with Key

What if $F(K, x) = K \oplus x \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0 \oplus K_1$$

$$L_2 = L_0 \oplus R_0 \oplus K_1$$

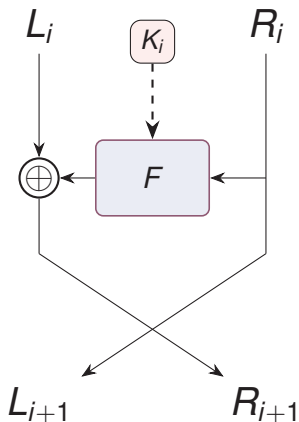
$$R_2 = R_0 \oplus (L_0 \oplus L_2) \oplus K_2$$

$$= L_0 \oplus K_1 \oplus K_2$$

Result

The output is just a linear function of the input and keys. Broken with basic algebra. No security!

Exploring Weak Functions



Case 3: Simple XOR with Key

What if $F(K, x) = K \oplus x \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0 \oplus K_1$$

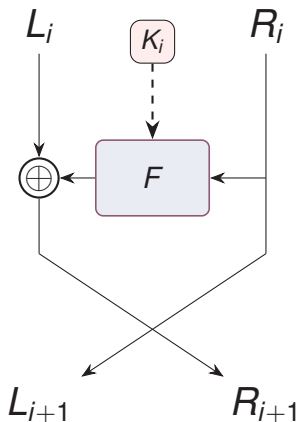
$$L_2 = L_0 \oplus R_0 \oplus K_1$$

$$\begin{aligned} R_2 &= R_0 \oplus (L_0 \oplus L_2) \oplus K_2 \\ &= L_0 \oplus K_1 \oplus K_2 \end{aligned}$$

Result

The output is just a linear function of the input and keys. Broken with basic algebra. No security!

Exploring Weak Functions



Case 3: Simple XOR with Key

What if $F(K, x) = K \oplus x \forall x$?

$$L_1 = R_0$$

$$R_1 = L_0 \oplus R_0 \oplus K_1$$

$$L_2 = L_0 \oplus R_0 \oplus K_1$$

$$R_2 = R_0 \oplus (L_0 \oplus L_2) \oplus K_2$$

$$= L_0 \oplus K_1 \oplus K_2$$

Result

The output is just a linear function of the input and keys. Broken with basic algebra. **No security!**

Conclusion: The Round Function is Everything

Main Lesson

A Feistel network is excellent **structure**, but its **round function** is everything.

- ▶ Weak or simple F functions lead to ciphers that are trivially broken.
- ▶ A strong F function must be **non-linear**

Conclusion: The Round Function is Everything

Main Lesson

A Feistel network is excellent **structure**, but its **round function** is everything.

- ▶ Weak or simple F functions lead to ciphers that are trivially broken.
- ▶ A strong F function must be **non-linear**

Conclusion: The Round Function is Everything

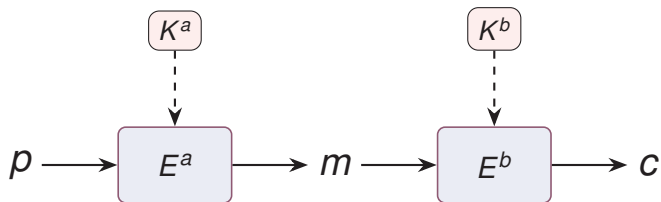
Main Lesson

A Feistel network is excellent **structure**, but its **round function** is everything.

- ▶ Weak or simple F functions lead to ciphers that are trivially broken.
- ▶ A strong F function must be **non-linear**

Meet-in-the-middle

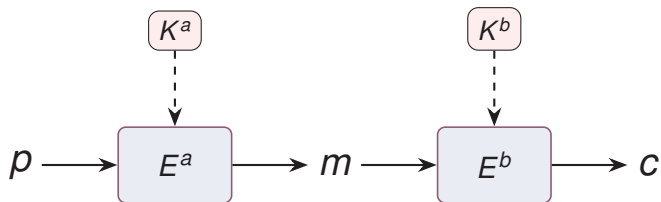
Double Encryption



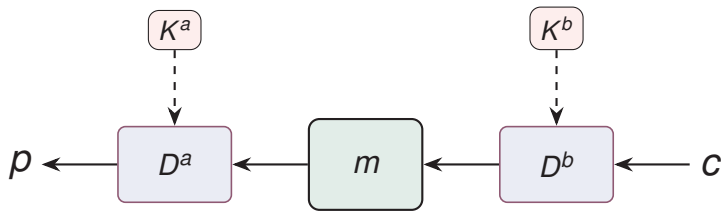
The Encryption Process: $c = E^b(K^b, \underbrace{E^a(K^a, p)}_m)$, where master-key $K = (K^a, K^b)$



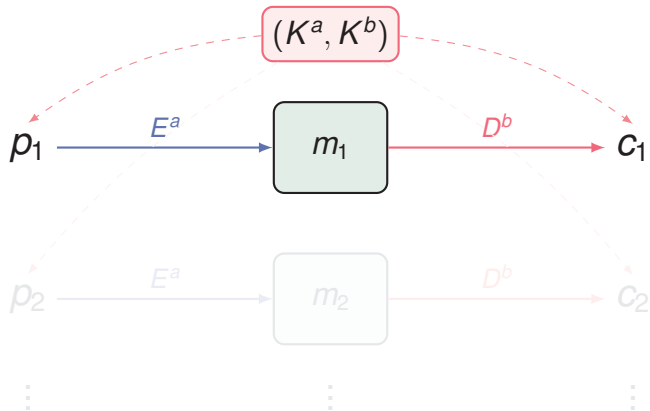
Double Encryption



The Encryption Process: $c = E^b(K^b, \underbrace{E^a(K^a, p)}_m)$, where master-key $K = (K^a, K^b)$

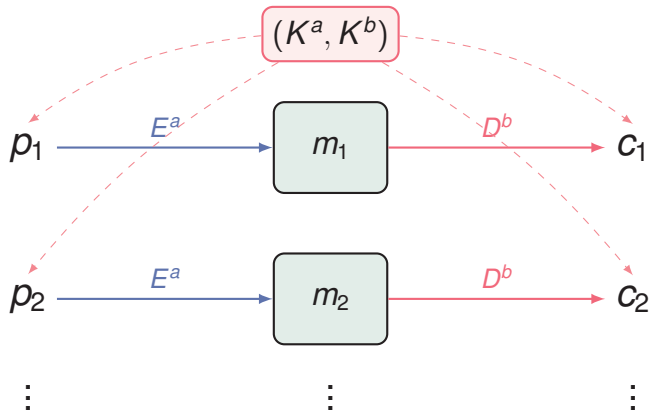


The Filtering Condition



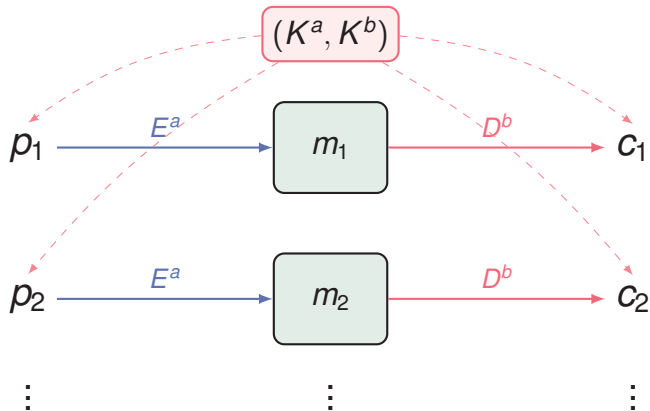
The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i)$ for all $i = 1, 2, \dots, t$

The Filtering Condition



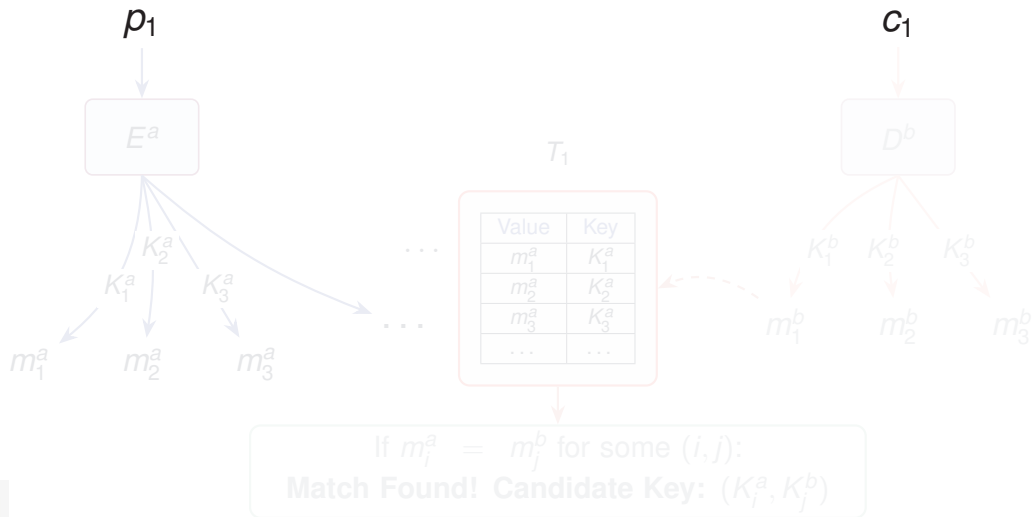
The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i)$ for all $i = 1, 2, \dots, t$

The Filtering Condition

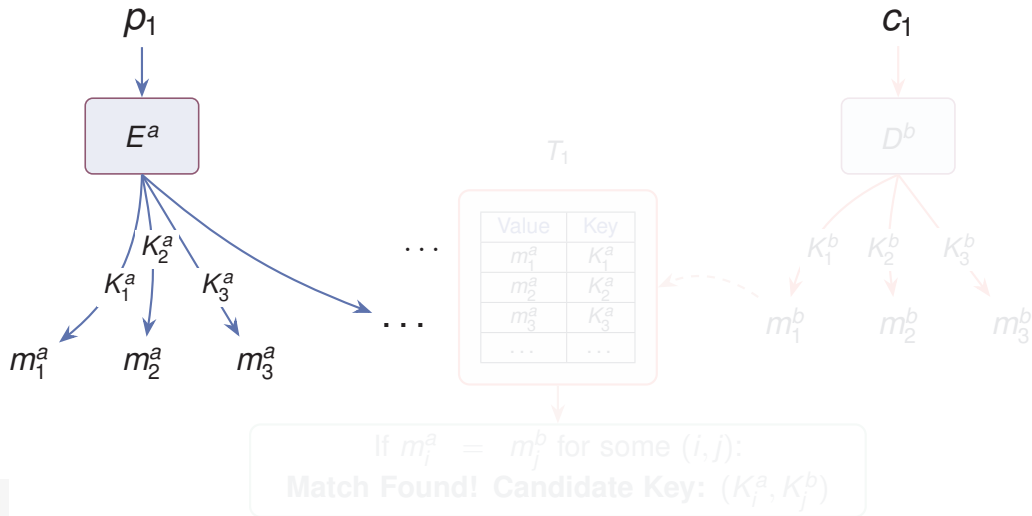


The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i)$ for all $i = 1, 2, \dots, t$

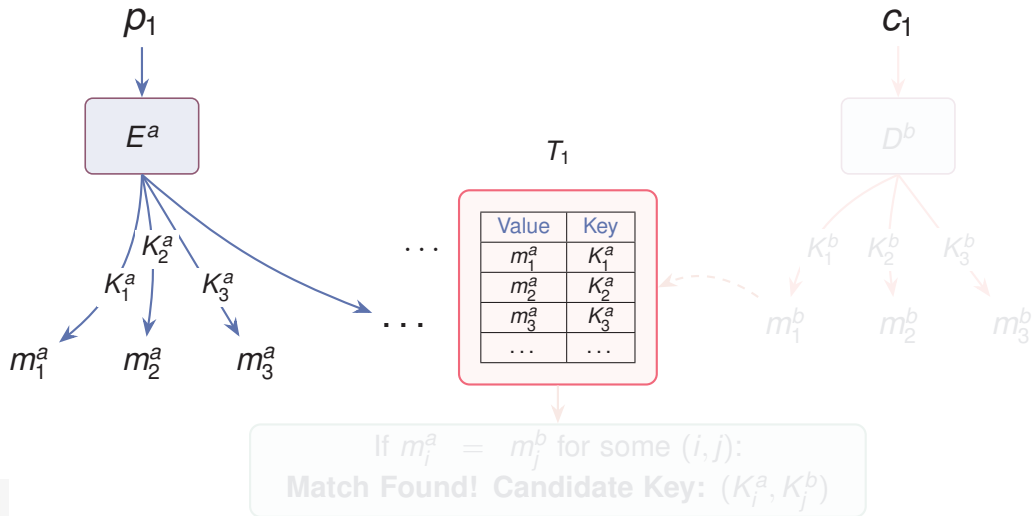
Meet-in-the-Middle Attack



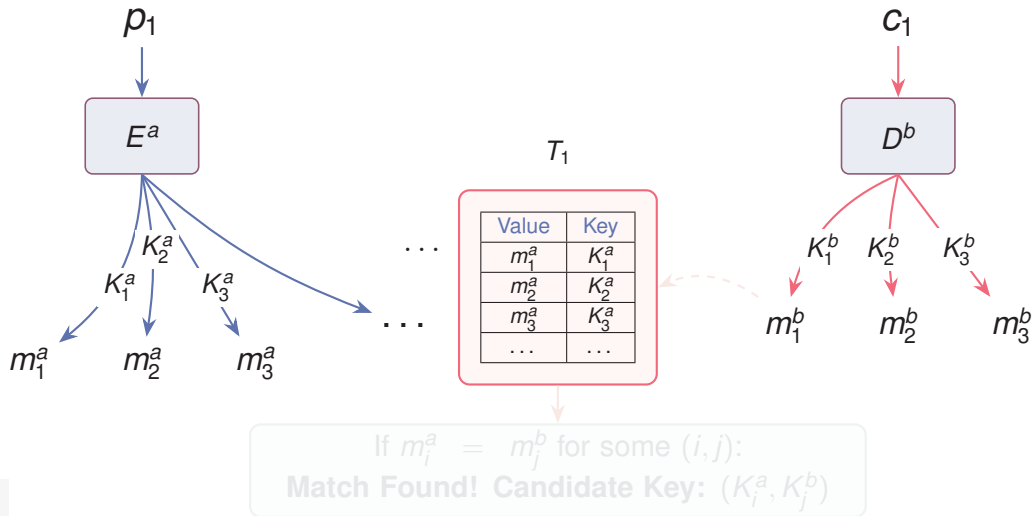
Meet-in-the-Middle Attack



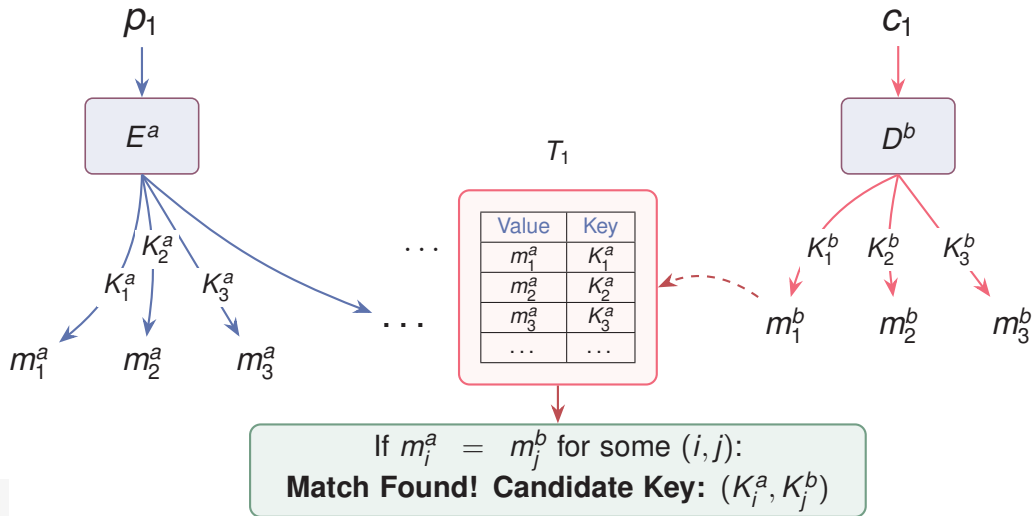
Meet-in-the-Middle Attack



Meet-in-the-Middle Attack



Meet-in-the-Middle Attack



Meet-in-the-middle (cont.)

Consider a plaintext-ciphertext pair (p_1, c_1) .

1. Construct a table T_1 with rows of the form (m, K^a) , where $E^a(K^a, p_1) = m$
2. Sort the table T_1 by first element
3. For every K^b , search for $D^b(K^b, c_1)$ in T_1 .
4. **Match** $\implies$ we have a candidate key $K = (K^a, K^b)$

Complexity of the attack

Time complexity: $O(n2^n)$ and Memory complexity: $O(n2^n)$

1. 2^n : Compute E^a for every K^a
2. $n2^n$: To sort the table T_1 .
3. $2^n(n + n)$: Size of the table T_1

Meet-in-the-middle (cont.)

Consider a plaintext-ciphertext pair (p_1, c_1) .

1. Construct a table T_1 with rows of the form (m, K^a) , where $E^a(K^a, p_1) = m$
2. Sort the table T_1 by first element
3. For every K^b , search for $D^b(K^b, c_1)$ in T_1 .
4. **Match** $\implies$ we have a candidate key $K = (K^a, K^b)$

Complexity of the attack

Time complexity: $O(n2^n)$ and Memory complexity: $O(n2^n)$

1. 2^n : Compute E^a for every K^a
2. $n2^n$: To sort the table T_1 .
3. $2^n(n + n)$: Size of the table T_1

Meet-in-the-middle (Success Probability)

1. The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
2. What is the expected number of keys (K^a, K^b) s.t.
 - ▷ $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
3. For fixed p and a random K^a , $\Pr(E^a(K^a, p) = z) = 2^{-\kappa}$ for any fixed z
4. Similarly, for fixed c and a random K^b $\Pr(D^b(K^b, c) = z) = 2^{-\kappa}$ for any z
5. Thus $\Pr(E^a(K^a, p) = D^b(K^b, c)) = 2^{-\kappa}$.
6. The above equation holds for all t pairs has probability $2^{-t\kappa}$.
7. As there are total $2^{2\kappa}$ possible keys, the expected number of keys is $2^{2\kappa-t\kappa}$.

Modified DES

Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16.

Meet-in-the-middle (Success Probability)

1. The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
2. What is the expected number of keys (K^a, K^b) s.t.
 - ▷ $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
3. For fixed p and a random K^a , $\Pr(E^a(K^a, p) = z) = 2^{-\kappa}$ for any fixed z
4. Similarly, for fixed c and a random K^b $\Pr(D^b(K^b, c) = z) = 2^{-\kappa}$ for any z
5. Thus $\Pr(E^a(K^a, p) = D^b(K^b, c)) = 2^{-\kappa}$.
6. The above equation holds for all t pairs has probability $2^{-t\kappa}$.
7. As there are total $2^{2\kappa}$ possible keys, the expected number of keys is $2^{2\kappa-t\kappa}$.

Modified DES

Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16.

Meet-in-the-middle (Success Probability)

1. The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
2. What is the expected number of keys (K^a, K^b) s.t.
 - ▷ $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
3. For fixed p and a random K^a , $\Pr(E_a(K^a, p) = z) = 2^{-\kappa}$ for any fixed z
4. Similarly, for fixed c and a random K^b $\Pr(D^b(K^b, c) = z) = 2^{-\kappa}$ for any z
5. Thus $\Pr(E^a(K^a, p) = D^b(K^b, c)) = 2^{-\kappa}$.
6. The above equation holds for all t pairs has probability $2^{-t\kappa}$.
7. As there are total $2^{2\kappa}$ possible keys, the expected number of keys is $2^{2\kappa-t\kappa}$.

Modified DES

Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16.

Meet-in-the-middle (Success Probability)

1. The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
2. What is the expected number of keys (K^a, K^b) s.t.
 - ▷ $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
3. For fixed p and a random K^a , $\Pr(E_a(K^a, p) = z) = 2^{-\kappa}$ for any fixed z
4. Similarly, for fixed c and a random K^b $\Pr(D^b(K^b, c) = z) = 2^{-\kappa}$ for any z
5. Thus $\Pr(E^a(K^a, p) = D^b(K^b, c)) = 2^{-\kappa}$.
6. The above equation holds for all t pairs has probability $2^{-t\kappa}$.
7. As there are total $2^{2\kappa}$ possible keys, the expected number of keys is $2^{2\kappa-t\kappa}$.

Modified DES

Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16.

Meet-in-the-middle (Success Probability)

1. The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
2. What is the expected number of keys (K^a, K^b) s.t.
 - ▷ $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
3. For fixed p and a random K^a , $\Pr(E_a(K^a, p) = z) = 2^{-\kappa}$ for any fixed z
4. Similarly, for fixed c and a random K^b $\Pr(D^b(K^b, c) = z) = 2^{-\kappa}$ for any z
5. Thus $\Pr(E^a(K^a, p) = D^b(K^b, c)) = 2^{-\kappa}$.
6. The above equation holds for all t pairs has probability $2^{-t\kappa}$.
7. As there are total $2^{2\kappa}$ possible keys, the expected number of keys is $2^{2\kappa-t\kappa}$.

Modified DES

Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16.

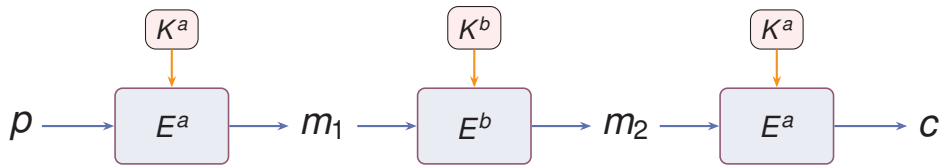
Meet-in-the-middle (Success Probability)

1. The Constraint: $E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
2. What is the expected number of keys (K^a, K^b) s.t.
 $\triangleright E^a(K^a, p_i) = m_i = D^b(K^b, c_i), \quad \forall i = 1, 2, \dots, t$
3. For fixed p and a random K^a , $\Pr(E_a(K^a, p) = z) = 2^{-\kappa}$ for any fixed z
4. Similarly, for fixed c and a random K^b $\Pr(D^b(K^b, c) = z) = 2^{-\kappa}$ for any z
5. Thus $\Pr(E^a(K^a, p) = D^b(K^b, c)) = 2^{-\kappa}$.
6. The above equation holds for all t pairs has probability $2^{-t\kappa}$.
7. As there are total $2^{2\kappa}$ possible keys, the expected number of keys is $2^{2\kappa-t\kappa}$.

Modified DES

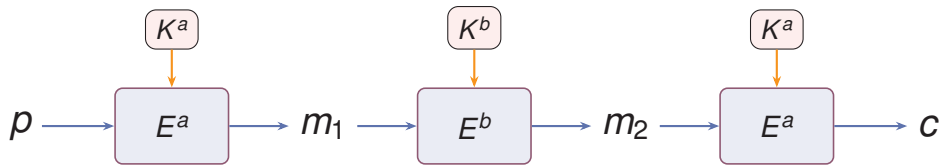
Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16.

Meet-in-the-middle Attack on Triple Encryption



1. Suppose for any (m_1, m_2) , it is possible to find in constant time all keys K such that $m_2 = E^b(K, m_1)$
 - 1.1 Can you recover the full key of K^a, K^b using three plaintext-ciphertext pair?
 - 1.2 What is the time complexity?

Meet-in-the-middle Attack on Triple Encryption



1. Suppose for any (m_1, m_2) , it is possible to find in constant time all keys K such that $m_2 = E^b(K, m_1)$
 - 1.1 Can you recover the full key of K^a, K^b using three plaintext-ciphertext pair?
 - 1.2 What is the time complexity?

Attack on Triple Encryption

1. Let $(p_1, c_1), (p_2, c_2), (p_3, c_3)$ be three collected plaintext-ciphertext pairs.
2. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:
 - 2.1 Compute $m_1 = E^a(K^a, p_1)$ and $m_2 = D^a(K^a, c_1)$
 - 2.2 Find the set S of all keys K^b such that $m_2 = E^b(K^b, m_1)$ (it takes constant time).
 - 2.3 For every $K^b \in S$: if $c_2 = E_{K^a, K^b}(p_2)$ and $c_3 = F_{K^a, K^b}(p_3)$ then
 $K_g = K_g \cup \{(K^a, K^b)\}$.

Complexity of the attack

1. The outer loop runs for 2^n
2. Running time of the inner loop is linear in $|S|$
3. What is the probability that an incorrect key will be in K_g ?
4. We expect that $|S| \approx 1$, so total running time is 2^n

Attack on Triple Encryption

1. Let $(p_1, c_1), (p_2, c_2), (p_3, c_3)$ be three collected plaintext-ciphertext pairs.
2. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:
 - 2.1 Compute $m_1 = E^a(K^a, p_1)$ and $m_2 = D^a(K^a, c_1)$
 - 2.2 Find the set S of all keys K^b such that $m_2 = E^b(K^b, m_1)$ (it takes constant time).
 - 2.3 For every $K^b \in S$: if $c_2 = E_{K^a, K^b}(p_2)$ and $c_3 = F_{K^a, K^b}(p_3)$ then
 $K_g = K_g \cup \{(K^a, K^b)\}$.

Complexity of the attack

3. What is the probability that an incorrect key will be in K_g ?
4. We expect that $|S| \approx 1$, so total running time is 2^n

Attack on Triple Encryption

1. Let $(p_1, c_1), (p_2, c_2), (p_3, c_3)$ be three collected plaintext-ciphertext pairs.
2. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:
 - 2.1 Compute $m_1 = E^a(K^a, p_1)$ and $m_2 = D^a(K^a, c_1)$
 - 2.2 Find the set S of all keys K^b such that $m_2 = E^b(K^b, m_1)$ (it takes constant time).
 - 2.3 For every $K^b \in S$: if $c_2 = E_{K^a, K^b}(p_2)$ and $c_3 = F_{K^a, K^b}(p_3)$ then
 $K_g = K_g \cup \{(K^a, K^b)\}$.

Complexity of the attack

3. What is the probability that an incorrect key will be in K_g ?
4. We expect that $|S| \approx 1$, so total running time is 2^n

Attack on Triple Encryption

1. Let $(p_1, c_1), (p_2, c_2), (p_3, c_3)$ be three collected plaintext-ciphertext pairs.
2. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:
 - 2.1 Compute $m_1 = E^a(K^a, p_1)$ and $m_2 = D^a(K^a, c_1)$
 - 2.2 Find the set S of all keys K^b such that $m_2 = E^b(K^b, m_1)$ (it takes constant time).
 - 2.3 For every $K^b \in S$: if $c_2 = E_{K^a, K^b}(p_2)$ and $c_3 = F_{K^a, K^b}(p_3)$ then
 $K_g = K_g \cup \{(K^a, K^b)\}$.

Complexity of the attack

1. The outer loop runs for 2^κ
2. Running time of the inner loop is linear in $|S|$
3. What is the probability that an incorrect key will be in K_g ?
4. We expect that $|S| \approx 1$, so total running time is 2^n

Attack on Triple Encryption

1. Let $(p_1, c_1), (p_2, c_2), (p_3, c_3)$ be three collected plaintext-ciphertext pairs.
2. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:
 - 2.1 Compute $m_1 = E^a(K^a, p_1)$ and $m_2 = D^a(K^a, c_1)$
 - 2.2 Find the set S of all keys K^b such that $m_2 = E^b(K^b, m_1)$ (it takes constant time).
 - 2.3 For every $K^b \in S$: if $c_2 = E_{K^a, K^b}(p_2)$ and $c_3 = F_{K^a, K^b}(p_3)$ then
 $K_g = K_g \cup \{(K^a, K^b)\}$.

Complexity of the attack

1. The outer loop runs for 2^κ
2. Running time of the inner loop is linear in $|S|$
3. What is the probability that an incorrect key will be in K_g ?
4. We expect that $|S| \approx 1$, so total running time is 2^n

Attack on Triple Encryption

Construct T containing rows of the form (m_2, k_2) where $m_2 = E^b(K^b, 0^n)$ and sort the table by m_2 . (Preprocessing time: $n2^n$) First collect two plaintext-ciphertext pairs $(p_2, c_2), (p_3, c_3)$. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:

1. Compute $p_1 = D^a(K^a, 0^n)$ and ask for an encryption of p_1 to F_{K^a, K^b}
2. Let the result be c_1 and $m_2 = f_{K^a}^{-1}(c_1)$
3. From table T , we find the row with first element m_2
4. If the second element is K^b , check other two pairs of data with (K^a, K^b)

What is the complexity of the whole attack?

Attack on Triple Encryption

Construct T containing rows of the form (m_2, k_2) where $m_2 = E^b(K^b, 0^n)$ and sort the table by m_2 . (Preprocessing time: $n2^n$) First collect two plaintext-ciphertext pairs $(p_2, c_2), (p_3, c_3)$. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:

1. Compute $p_1 = D^a(K^a, 0^n)$ and ask for an encryption of p_1 to F_{K^a, K^b}
2. Let the result be c_1 and $m_2 = f_{K^a}^{-1}(c_1)$
3. From table T , we find the row with first element m_2
4. If the second element is K^b , check other two pairs of data with (K^a, K^b)

What is the complexity of the whole attack?

Attack on Triple Encryption

Construct T containing rows of the form (m_2, k_2) where $m_2 = E^b(K^b, 0^n)$ and sort the table by m_2 . (Preprocessing time: $n2^n$) First collect two plaintext-ciphertext pairs $(p_2, c_2), (p_3, c_3)$. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:

1. Compute $p_1 = D^a(K^a, 0^n)$ and ask for an encryption of p_1 to F_{K^a, K^b}
2. Let the result be c_1 and $m_2 = f_{K^a}^{-1}(c_1)$
3. From table T , we find the row with first element m_2
4. If the second element is K^b , check other two pairs of data with (K^a, K^b)

What is the complexity of the whole attack?

Attack on Triple Encryption

Construct T containing rows of the form (m_2, k_2) where $m_2 = E^b(K^b, 0^n)$ and sort the table by m_2 . (Preprocessing time: $n2^n$) First collect two plaintext-ciphertext pairs $(p_2, c_2), (p_3, c_3)$. Initialize a set $K_g = \phi$. Then for every key K^a do the followings:

1. Compute $p_1 = D^a(K^a, 0^n)$ and ask for an encryption of p_1 to F_{K^a, K^b}
2. Let the result be c_1 and $m_2 = f_{K^a}^{-1}(c_1)$
3. From table T , we find the row with first element m_2
4. If the second element is K^b , check other two pairs of data with (K^a, K^b)

What is the complexity of the whole attack?

Thank You for your attention!
Any questions?